Speculating with the body: Imagining designs for women's embodied empowerment within the practice of feminist self-defense

Bojana, BN, Nikolovska

Uppsala University, Department of Informatics and Media, b.b.nikolovska@gmail.com

Feminist self-defense is a form of victim prevention training with a plethora of positive physical, mental, and social outcomes. In Human-Computer Interaction (HCI) self-defense is still a relatively under-researched topic. As such, inspired by HCI's recent interest in feminist causes and corporeal practices, the goal of the study is to explore how embodied interaction design can empower beginner self-defense practitioners. To do so, the study was conducted via two methods: semi-structured interviews with students and teachers, and a participatory speculative design workshop with novice practitioners. This resulted in several speculative design concepts based on the use of felt experiences as a design resource. The concepts demonstrate how design can be used as a vehicle for imagining feminist technology that challenges gender norms and plays the role of scaffolding for cultivating embodied empowerment.

CCS CONCEPTS • Human-Centered Computing • Interaction Design • Interaction Design Process and Methods • Participatory Design

Additional Keywords and Phrases: Embodied Interaction Design, Speculative Design, Soma Design

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1 INTRODUCTION

Feminist self-defense is a form of victim prevention that has a profound impact on women's sense of self and their being-in-the-world [51]. When the topic of women's self-defense is brought up, it is often associated with images of women using pepper spray against an unknown assailant. It is also not uncommon for the practice to be equated solely to the teachings of techniques on how to physically defend oneself from an attack [49]. Indeed, physical resistance is part of the training, however, feminist self-defense goes beyond this. It engages with the topics of mental strength, women's empowerment, the relationship between women and their bodies, and the experience of inhabiting a female-gendered body [44]. As such, feminist self-defense can be seen as a corporeal practice grounded in feminist theory, that centers the lived body [49]. It can also be defined as a bridge between two forms of feminist activism, empowerment via building of physical strength (predominantly focused on the body) and feminist consciousness-raising (predominantly focused on the mind), in order to enact an embodied empowerment [46].

As such, the focus of the project is on exploring the embodied experiences of self-defense practitioners. It seeks to answer the question of *how embodied interaction design can empower beginner practitioners during their engagement with the practice of feminist self-defense.* To do so, I inquired into the bodily sensations and emotions that arise for first-time practitioners during a self-defense class and explored how they can be used as a design resource. The project's outcome is several speculative concepts developed during a participatory design workshop that point towards ways that technology can support practitioners. The concepts are envisioned as "entrance points for critical thinking" [27] and explorations of possible futures [19] of technology entering the space of feminist self-defense.

Recognizing the ambiguity of the term empowerment, I choose to adopt the definition of empowerment as a "process by which those who have been denied the ability to make strategic life choices acquire that ability by accessing material, information, and ideological resources" [48], whereby the teachings of feminist self-defense are seen as the resource that contributes toward an embodied empowerment - the process by which feminist self-defense practitioners gain the power to challenge the culture of violence against women that serves as a tool for bodily discipline [42].

Moreover, this project is situated at the intersection of feminist HCI [6] and design for bodily practices, and aims to contribute to furthering HCI's potential for designing for liberation [29].

2 BACKGROUND

2.1 Feminist Self-Defense

Violence against women is a complex social and political issue that in order to be tackled requires large-scale social changes [13, 51]. However, there is not one single course of action that can be charted to bring about a systemic change [26], instead, multiple initiatives serve as pieces of a larger puzzle [26]. Noted examples of such initiatives are support centers for survivors of sexual violence [44], violence prevention programs that focus on potential attackers or bystanders [26], and feminist self-defense training [26] - the focus of this project.

Feminist self-defense is as a form of victim prevention that teaches women how to prevent and respond to violence [26, 30, 51]. Its contemporary form stems from the activism of second-wave feminism [34, 44], though its practice has also been present during the suffragette movement [50]. Even though it is an individual practice, it has been argued that it has wider political and social consequences [13]. Often framed as a tool for fighting rape culture and the effects of gender ideology [13], the goals of feminist self-defense classes are to train and support women in acquiring an embodied form of resistance and to overcome normative beliefs about gender, violence, and the body [13]. This in turn leads to the creation of a "new soma reality" - the acquiring of a new bodily knowledge that contradicts the notion that a female-presenting body is weaker than a male-presenting body, which in turn affects women's being-in-the-world [44].

2.1.1 The "feminist" in feminist self-defense

Not all women's self-defense programs can be labeled as feminist self-defense. Even though they may strive to teach protection from violence, in doing so some also perpetuate harmful gender stereotypes [26]. Non-feminist self-defense programs are those that frame resistance as avoidance of certain places or activities, or as reliance on external agents for protection, such as male presenting companions [51]. This type of advice has historically been the most common and has the effect of constricting women's ability to be an independent actor in society and to partake in a plethora of experiences [13, 26, 51]. Other programs may not shy away from the physical aspect of self-defense, such as those inspired by martial arts, yet they can not necessarily be classified as feminist either [52]. Namely, they have been critiqued for marketing themselves as self-defense classes for women, without understanding and centering the needs of their target audience [13,

51]. These classes focus on the teaching of stylistic techniques that may take years to master and as such may not be as accessible [13, 51]. Moreover, as the classes may be co-ed, they may fail to facilitate the creation of a safe space [13, 51]. Some programs have also been found to employ victim-blaming language that frames assault as an outcome of women's personal choices [13].

What differentiates feminist self-defense from these examples is its grounding in feminist theory, the techniques that are taught, the context of the class, and the underlying power structures [51]. Namely, these classes usually take place in women-only groups. Here the trainers are not positioned as teachers in a traditional martial arts sense, but as "supporters" of the practitioners [42]. They acknowledge and respect the situated knowledge that different participants may bring to the training [42]. Furthermore, feminist self-defense builds on the understanding that the difference between female and male bodies is not natural, but socially constructed. Physical properties believed to be inherently female such as gentleness, lower levels of physical strength, and non-aggression are seen as a manifestation of "sexed embodiment" - the effect of gender ideology shaping the body and bodily practices into what it prescribes to be the natural form. As such, feminist self-defense seeks to actively undo these beliefs, to show that women's bodies are not inherently weaker, and that they are indeed capable of active self-defense. It disrupts the belief that men's bodies are invulnerable, inherently stronger, and that protection against an attack is futile. [44] It does this via teachings that aim to empower women to feel and perceive their bodies as capable and worth protecting [13, 30]. Feminist self-defense classes also undermine rape culture by framing sexual violence as a social problem as opposed to an individual issue. They emphasize that the victim is never responsible for an assault, and as such touch upon issues of victim-blaming and self-blame. The training recognizes that violence can manifest in many forms and be perpetrated by strangers, acquaintances, or intimate partners, thus fighting long-standing assault myths. [13, 26]

2.1.2 The elements of a feminist self-defense program

The elements of a feminist self-defense class can be roughly divided into the physical and the psychological. Although this is a loose separation as they are actively interlinked, and in the messy reality of a defense class they can often overlap and influence each other. For example, to achieve mastery of techniques, one should also address any emotional barriers that may hinder this achievement [42].

Feminist self-defense teaches that resistance can be "forceful physical resistance (e.g. fighting the attacker), non-forceful physical resistance (e.g. running away), forceful verbal resistance (e.g. yelling), non-forceful verbal resistance (e.g. pleading)" [30]. The techniques should not be seen as prescriptions of how a woman should react, but as possible courses of action, as ultimately it is a woman's decision how she should act (or not act) in the case of an assault [13]. These techniques are usually relatively quick to master [26]. They can be practiced during mock scenarios of simulated assaults against padded attackers, which can vary in their degree of volatility [34], against instructors who hold pads [30] or against other students who roleplay an attacker [42]. [23] point to the importance of the group in mastering these techniques. Namely, the group acts as an emotional support tool and can help participants process trauma. Discussions are another important element of a class, as they allow participants to process experiences and engage in meaningful dialogue [52]. The use of self-defense tools is usually not present in most classes, as that requires relying on an external tool that may malfunction, instead of relying on one's own body [52]. [1]

1 It should be noted that not all of the listed elements are present in every program [30].

2.1.3 Outcomes and Effects

The intended outcome of feminist self-defense is the acquiring of skills to prevent and respond to violence. Research points that participants who have completed the training have higher rates of success in preventing an assault [12]. The existing body of literature also shows a plethora of transformative effects [26, 30, 44]. Namely, women who have taken feminist self-defense classes report having increased feelings of agency, self-worth, and confidence in their own bodies, whereby the body is experienced as powerful [30]. This reverberates in other aspects of women's lives, whereby women feel more assertive, able to establish boundaries, and communicate needs [30]. Recent research points out that even though it may take some time for the lessons to be "transferred to muscle memory" [34], once acquired they have stayed with women for decades, essentially becoming embedded in their bodily schema [59]. [2]

2.2 Technology for Self-Defense

In HCI, research on prevention of violence against women can be seen in studies that address intimate partner violence [45, 60], sexual harassment [1, 2, 16], risk assessment [15], as well as safety in public spaces [47], to name a few. However, feminist self-defense remains an under-researched topic. Currently, the engagement of technology with the subject of feminist self-defense is mostly visible in design of tools for self-defense. These tools can take the form of wearables, apps, or a combination of the two [65]. The devices are often framed as artifacts that enhance the female body with the property to defend itself [53]. However, they have also been critiqued for perpetuating the belief that women's bodies are inherently weak and for having a non-nuanced view of what constitutes an assault. These artifacts, coined by Shelby [2020] as "pink self-defense commodities", frame violence against women as an inevitable event that can be mitigated via personal risk management, as opposed to a systemic issue that needs to be eradicated. [8, 53, 63]. As such, these devices reposition self-defense from a bodily faculty to a feature of an artifact, thus impeding the creation of tacit knowledge [53]. This points to an opportunity for design to more meaningfully support the goals of feminist self-defense.

2.3 Theoretical Framing

2.3.1 About Bodies

"If bodies are objects or things, they are like no others, for they are the centers of perspective, insight, reflection, desire, agency" [28]. Designing and speculating with the practice of self-defense is to design and speculate with a body in movement. From a kinesthetic perspective, bodily movements are a sum of small muscle movements - executed via small parts of the body such as the facial muscles, the fingers, or the feet, and big muscle movements - executed via large parts of the body and refer to activities such as running, walking, and punching [22]. A body in movement also engages one's spatial awareness, namely the personal space[3], peripersonal space[4], and extra-personal space[5] [35]. Any audio, tactile, or olfactory stimuli can be seen as contributing factors to the experience.

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2 It should be noted that most of the existing research is based on the experiences of white women and is grounded in Western feminism, as such there have been calls for the incorporation of intersectionality and further exploring the lived experience and needs of non-white, non-Western, and differently abled women. [44, 56]

3 The space that is occupied by a body. [35]

4 The space within reach of the limbs. [35]

5 The space just beyond the reach of the limbs. [35]

To center this experience, the study is grounded in the theory of embodiment - a philosophical thought that refers to how via our bodies we are always rooted in the world. Namely, it is our physical, social, and cultural bodies that shape our experience in the world. In HCI, this view has been used as a way to explore the design of technology that can bridge the mind-body dualism and create artifacts that connect people with their bodies and environments, or what Dourish labels as embodied interaction design. [17] As such, theories of embodied interaction, such as embodied cognition [35], Merleau-Ponty's phenomenology of perception [58], and soma design [31] are relevant to this study.

According to Merleau-Ponty, our kinesthetic sense is what grounds us in the world [22, 58], while, according to Gibson, our repertoire of movements is what influences our perception of the world [35]. We can perceive and act on the affordances of the world due to the physical shape, properties, and skills of the body [35]. Thus, any change in our bodily skills will result in a change in how we perceive and act in the world [35]. However, the ways in which we move our bodies and the skills we train for is influenced by social values [35]. Gender - a set of socially constructed beliefs about bodily differences can be seen as one such influence. In a recent turn towards the corporeal, feminist theorists have postulated that gender ideals manifest themselves not only on a consciousness level, but on a bodily level as well. Namely, traditional feminine norms such as gentleness, nurturing, and non-aggression have a direct effect on how women move and hold their bodies. This is labeled as "feminine habitus" - a set of gestures that are influenced by idealized standards of femininity. [43]

Beauvoir suggested that due to social influences girls as compared to boys develop a different relationship to their bodies, which in turn influenced how they come to interact with the world [39]. "Women are not as likely as men to have developed a relationship with their bodies as agents, as instruments of action" [43]. This can render female gendered bodies as passive and docile[6]. Young [2005] further notes that women may be less likely to exhibit physical force in situations when that might be needed, not due to inherent physical weakness as compared to men, but due to the embodiment of gender ideology [39, 66]. She labels this as "inhibited intentionality", which influences a woman's sense of agency and power [39, 66]. Considering this, feminist self-defense can be seen as an interruption of sexed embodiment via the teaching of new bodily skills, which has a fundamental influence in how women act in the world.

2.3.2 Bodily Skills

The body's ability to remember practiced movements refers to kinesthetic memory. It is what allows us to learn new skills such as riding a bike or throwing a punch, and what leads to the creation of embodied knowledge [36]. Learning a new skill often implies the learning of how to execute an action, when to execute it, and being aware of how different environments may influence its execution [22]. In HCI, there is an increasing interest in how this can be supported by interaction design [22]. The development of such technology is based on the notion that design can support the creation of new skills and prompt new bodily experiences [35]. As such, new tools can lead to new ways of being in the world [35]. However, there is also an awareness that when any technology becomes part of a corporeal practice, it can be expected that it might influence the body's experience with said practice [41].

Knowing that a new skill has been successfully acquired is often determined not by how it looks when it is performed, but by how it feels [22]. Considering this, I looked towards soma design, as an approach that focuses on the felt experience of a bodily practice. [32] states that soma design is a valuable lens through which to explore bodily transformation, and the effects of feminist self-defense can certainly be seen as such. In addition, [41] write that the discussion of bodily habits,

6 The image of the female body as disconnected from any physicality and strength historically has not been applicable to all female gendered bodies and it mainly referred to Western, white, upper and middle class, heterosexual women. [43]

which one can argue is also part of feminist self-defense, can lead to a greater awareness and aesthetic appreciation. However, considering that self-defense can also bring about negative emotions, the study modifies the goals of soma design and adopts the lens not for aesthetic appreciation, but for embodied empowerment.

3 METHODOLOGY AND METHOD

3.1 Participatory Speculative Design

Given the focus on embodied experiences during a feminist critical practice, the study aligns itself with feminist epistemologies. The research thus takes the form of an exploratory design research informed by the feminist HCI qualities of pluralism, participation, and embodiment [6]. It adopts the view that knowledge is socially and culturally situated [6, 51] and based on feminist standpoint theory [6, 7] pursues knowledge that stems from the experiences of the lived and cultural body [6, 51]. It thus recognizes lived and felt experience as a form of knowledge [27, 35].

This stance informed the choice of method - participatory speculative design. As this is one of the first instances where technology enters the space of feminist self-defense, the goal of the study is not to provide a single technological solution, but a plethora of design concepts that open a new design space. The use of a participatory design (PD) approach centers the experiences of the practitioners and facilitates a "shared creativity"[7] and "shared authorship"[8] levels of participation [20]. PD's commitment to respecting the knowledge of non-designer stakeholders, viewing the designer as a facilitator and users as co-creators [54] aligns with the goals of feminist design research. On the other hand, speculative design (SD) is utilized to show the power of design artifacts to "challenge the status quo and encourage imagination" [25]. It allows for imaging what feminist technology could look like [48]. This aligns with the project's goal of imaging technologies that are informed by women's experiences and bodily sensations.

PD compliments SD since it opens the practice to non-designers [20]. "If speculating about the future liberates the imagination with potentially emancipatory implications, why should that task remain exclusively at the hands of designers?" [20]. On the other hand, SD complements PD as it can shift the focus from the design of commercial artifacts to the imagination of future possibilities [4] and "to-be" states [54]. As such, via this method selection, I hope to utilize design's power for imagining new ways of being [33].

3.2 Data Collection

3.2.1 Workshops

The study was conducted via two workshops with 6 female identifying participants selected via convenience sampling. The first workshop, which lasted two hours, was conducted in a martial arts studio by a self-defense teacher. It provided the participants with the opportunity to become acquainted with the basics of feminist self-defense. The workshop began with an introduction to the practice, followed by an overview of possible threat scenarios. The participants were then presented with possible courses of action in such situations. Forms of resistance that aim to decrease one's perceived vulnerability, such as the use of one's loud voice, the use of assertive body posture, and an assertive walk were taught first. The participants were then prompted to practice walking like a strong person and shouting "No" or "Stop" with a deep,

7 "Non-designers brainstorm with the designers before and/or during the design process." [20]

8 "Non-designers produce the speculative designs alongside the designers." [20]

confident voice. This was followed by lessons on how to release oneself from various forms of body-grabs. The participants then got the chance to practice these techniques in mock fights among each other, while taking turns in "playing the perpetrator". The workshop concluded with a discussion session. (The full outline can be found in Appendix H). The data from the class was collected via a video recording.

The second workshop invited the participants to imagine technologies that would support them in their self-defense practice. The workshop lasted for two hours and was divided into two parts. In the first part, inspired by the notion that descriptions of lived female experiences are a "valuable resource for feminist projects" [66], the participants were given the opportunity to reflect on their experience and externalize emotions and bodily sensations via body maps [7]. In the second part of the workshop, the participants were asked to build on their experiences and use them to imagine future technologies. Separated into smaller groups the participants engaged in a collaborative imaging of technology that "lies slightly out of our grasp" [18]. Inspired by [64], bodystorming was prompted by custom designed wearable probes. They were ambiguous items that can be engaged with and placed on the body in a variety of ways (Appendix F). It is up to the person that is manipulating the object to create its meaning. The goal of the props is to facilitate thinking through the body [24]. To further support the speculation and provide a form of guidance, each group was given a prompt inspired by the background research (Appendix G). Careful consideration was taken not to construct the prompts as problem briefs as not to trigger a "solutionist" approach [9]. To facilitate the creation and communication of the imagined technology, the participants were provided with a selection of crafting materials to enable thinking via prototyping [36]. The data was collected via a video recording of the presentation of the speculations. (The full workshop protocol is included in Appendix E.)

3.2.2 Interviews

During the project semi-structured interviews were conducted with both self-defense participants (Appendix B) and self-defense instructors (Appendix A), resulting in a total number of 8 interviews (2 with instructors and 6 with participants) ranging in duration from thirty minutes to one hour. All interviews were conducted remotely. The interviews with the instructors were conducted at the beginning of the project with the goal of better understanding the space of feminist self-defense. The participants were interviewed after their participation in the self-defense workshop with the goal to better understand their experiences and emotions. This research method was selected as it lent itself as suitable for an exploration of a complex phenomenon [38]. Moreover, semi structured interviews using open questions were used as it allowed for "digging deeper" into the subject matter [38].

3.3 Data Analysis

The data from the interviews was transcribed verbatim with some mention of non-verbal communication [11]. The authenticity of the transcriptions was then compared to the original recordings. The data was subject to an inductive thematic analysis [11]. It was analyzed via two rounds of coding. The subsequent generated codes were grouped into two levels of subthemes, which were then grouped into 6 overarching themes conceptualized as "meaning-united stories" [10]. The data from the design workshop - the proposed speculative designs were grouped into themes based on their features and goals. Validity of the analysis is ensured by maintaining transparency about the execution of each step of the process.

3.4 Limitations

Due to the nature of the research design several limitations were encountered. Namely, the convenience sampling strategy resulted in a participant group that skewed towards a similar demographic (age: early 20s - mid 30s) and educational background (HCI).

As such the study may have only partially achieved a plurality of voices and experiences. Reflecting back, more effort could have been placed into recruiting participants outside of the field of HCI. This in turn could have led to a more varied set of design speculations. Furthermore, as the development of self-defense skills may take time to develop, the result may not be indicative of the experiences of practitioners who have undergone longer training sessions. Similarly, it can also be assumed that the short amount of time that the participants engaged in the practice will not be sufficient for the development of kinesthetic memory [36], which in turn may have influenced both the speculations and the interview data.

The selection of interviews as a data collection method naturally resulted in issues of participants relying on their memory of a self-defense class. The interviews were conducted separately from the context [38] with a gap of several days between the participation in the workshop and the participation in the interviews. This may have resulted in loss of capturing some aspects of the experience. However, it did provide a glimpse into how participants have begun to implement the lessons from the workshop. Moreover, the interview data may also be subject to bias, as knowing that the self-defense session was organized by the researcher may have resulted in hesitation of sharing any negative or critical impressions. A further possible limitation could also lie in the speculative artifacts as they currently only exist as concepts, as opposed to material artifacts. This may limit their potential in triggering the imagination [62].

3.5 Ethical Considerations

Considering that this might be the first time that participants have "seriously thought about the reality of being assaulted" [42], as well as the fact that the workshop activities may trigger negative emotions due to past trauma, special consideration was taken to ensure that there is a safe and supportive environment and for space to be made for any feelings that may arise [55]. Participants could opt out of engaging in certain activities or in the study altogether. Consent for data usage could be revoked at any time. Furthermore, considering that sensitive and private personal information may be shared, the video recordings were kept on an external hard drive, all data collected during the study was anonymized, and deleted after the project's competition. The consent forms for the self-defense workshop and the design workshop are in Appendix C and D, respectively. The consent for participation in the interviews and the participant's rights were shared verbally and recorded via audio.

4 RESULTS AND ANALYSIS

The results of the study are presented in two parts. The first part provides an overview of the participants' felt experiences of a self-defense class and discusses their role as a resource for design. The second part provides an overview of the participants' speculations regarding the role of future technology in supporting their practice.

4.1 Experiences of Feminist Self-Defense

4.1.1 Performative vs. Embodied Strength

The conversations with the participants point to the existence of conflicting notions over emotional and physical strength. Namely, strength was being perceived as a faculty the participants possess. However, they also mention a need to engage in a performance in which they actively communicate said possession via body language, in order to be seen as less vulnerable.

Feelings of strength usually manifested as trust in oneself that one is indeed able to persevere in the face of negative experiences.

> "There is a solution to everything. Like that was one of the things that I got subconsciously as a lesson from the workshop. So even if you find yourself having to take maybe a dark path to home, there is still like a solution…"

Strength was experienced both as gained from the participation in the class and as a pre-existing faculty that just needed to be uncovered. The sources of strength were usually located in either one's mind or body. Some participants identified confidence or mental strength as the foundation for being able to defend oneself, with strength described as something that can originate from within.

> "…once you are confident with yourself you can manage your body language and your next move…"

Others reflected on feelings of strength as being generated from movement or from the knowledge that one can indeed throw a punch.

> "…I felt strong that I can do stuff…"

This was also reflected in the conversations with the instructors, whereby they encourage students to practice body postures that elicit feelings of strength. It is interesting to note that despite showing awareness of how one's physicality can influence one's mental state, and vice-versa, there is still a notable separation between mind and body in the language used by both the participants and the instructors, with one almost always seen as more dominant than the other.

On the other hand, despite mentions of feeling strong, strength was also seen as a performance, with the exercises in assertive body posture seen as exercises in playing a role. In these instances, the body was perceived as a tool in making the act more convincing. Outside of the self-defense class, the body (and its posture) was experienced as a means for communicating one's strength to the environment. Taking on an assertive posture and walking confidently was experienced as an armor one could activate. However, its activation is entirely dependent on conscious thought and deliberate action.

> "…If you really feel uncomfortable in a situation, I think I would still go back to it…I think I would try to remind myself to be like, "OK, stand up straight and be more upright and not just slouch" … As a preventative measure…"

Similarly, one participant expressed a heightened awareness of how taking up more physical space can be a way of being in control of how they are approached in public. Implications of the conflict between "real" and "performed" strength can be seen in the proposed speculative designs. Technology such as a VR avatar that shows you an altered image of yourself as a "strong person" or an implant that alters your personality so you become a "strong person" can be seen as a way to bridge the gap between the performed and the embodied.

4.1.2 Negotiating Vulnerability

The self-defense class was also seen as a prompt for reflecting on one's vulnerability. The participants often examined how one's habits and behavior might be perceived as sources of vulnerability. They also pointed to events where they have an acute feeling of their vulnerability being evaluated.

> "… sometimes I'm more cautious because I feel like they just look at me and assess that I am alone and that is uncomfortable and I'm more actively thinking about what we were told …"

There was a prevailing notion of one's perception of vulnerability as something that can and should be managed, as being perceived as vulnerable was seen as a security risk. One's body posture and overall attitude play a significant role in this. As such, perceived vulnerability and preformed strength can be seen as two sides of the same coin.

> "…if you make yourself look like you are very vulnerable then you are most likely to be attacked or harassed…"

This was also reflected in the conversations with the instructors. Namely, a potential perpetrator should not get the impression that one is an easy target.

Yet, feelings of vulnerability were not seen as a constant part of women's lives, as most reported that they feel safe in general. This was attributed to one's environment or one's behavior. However, the self-defense class was also seen as a reminder that despite the feelings of safety in one's day-to-day life, due to the nature of existing in the world as a woman, one might be subject to violence.

> "…She [the instructor] was talking about things that we usually don't talk about. At least now we think…that everything is very safe around us and we try to do everything. But she was actually talking about things … That is important, and that can happen…"

Thus "feeling safe" is an unstable state that can be interrupted by an external factor. This can be seen as continually existing on a spectrum of possible victimhood, where one has to negotiate feelings of relative personal safety with systemic vulnerability due to having a female gendered body. This perception can be connected to rape culture, where violence is almost always seen as a possibility [43]. For the beginner self-defense practitioners, despite the (newly gained) awareness that they can have control over the possibility of victimization - "interrupting the script of an assault" [43], it seems like they still experienced their bodies as potential "victim bodies" [43]. As a result, participants often took a closer look at their behaviors and reflected on possible lifestyle changes that can reduce one's potential vulnerability.

4.1.3 Perceived Separation of Practice and Reality

From the conversation with the participants one can derive that the environment of the self-defense workshop was seen as a safe place, but also as removed from "real life". This contributed to how the lessons were experienced. Namely, participants reported not being in the right mindset to fully immerse themselves in the shouting exercises, the imagined attack exercise, and the mock fighting exercises. The lack of full immersion was attributed to not being in a fighting mindset, the need for a trigger that will elicit combative behavior, or due to experiencing awkwardness and thus hesitating to be fully physically invested.

> "…Because we were in a friendly environment and we were like laughing… I mean the scenario was like that. And I think that a thing like this [shouting exercise] ...it should come from here [points to chest]. I can only do that exercise, or reach to that point when I seriously feel something inside…"

During the sparring exercises, it was not uncommon to restrain strength, or to apologize after hitting one's partner. This can be seen as further contributing to the feelings of separation between mock fighting and "real" fighting.

> "…It was a lot of focus on apologizing and not grabbing too hard and you couldn't really put yourself into the role or the situation and really focus on "Oh, I'm being grabbed!"

However, this can also be seen as a reaction to transgressing the feminine habitus. The conversations with the instructors point out that the initial hurdle of becoming comfortable with shouting and fighting, might be the biggest obstacle to overcome.

Some participants also questioned if self-defense is something that can even be simulated, as one cannot know how they might feel or react during an actual attack. The felt discrepancy between the "real" and the "simulated" can also be found in the speculative designs. Namely, in the form of technology such as a simulated environment that goes beyond VR and incorporates bodily sensations. This allows one to experience an attack without being in actual danger.

4.1.4 A Self-Defense Class as a Space for Confronting One's Physicality

The class was experienced as a place where the participants could engage with the body in ways that are outside their movement repertoire. This can take the form of being prompted to shout, to hit, to experience what it feels to be pushed and to lose and regain balance.

> "…Just trying out these new things that you haven't really done before. I mean I've never grabbed someone like that before, so it was kind of actually fun…"

During the workshop participants came to face the implications of making such novel movements. This ranged from feelings of awkwardness to deriving pleasure from learning fighting maneuvers. In addition, the element of physicality helped to gain a better understanding of the verbal lessons shared during the class. However, the experience also prompted a reflection on the possible discrepancy between understanding what needs to be done during a certain exercise and the execution of said exercise on a bodily level.

> "…although she [the instructor] explained it, I wasn't quite sure how to let my body do what I wanted it to do…"

As such, some participants reflected on often wondering if they had correctly understood and executed a movement. Participants also expressed concern over the ability of screen-based technology to facilitate this aspect of the class. As such, during the speculative design workshop, the focus on engaging with one's physicality was most notably reflected in the form of a robot that takes the role of a fighting partner, measures how hard one hits, and encourages the use of more force if needed. In addition, a room where one is prompted to push bodily limits, as well as reflect on one's actions, can also be seen as a way to further explore one's physicality and go beyond one's movement repertoire.

4.1.5 The Group as a Mediator of the Experience

The attitude and approach of both the instructor and the other participants were seen as key factors contributing to how the self-defense class was experienced. Namely, the participants felt that they were in a safe space due to feeling comfortable in the group, and due to the friendly approach of the self-defense teacher. These sentiments were especially prominent when describing the overall atmosphere and the discussion session at the end of the workshop.

> "It was good that everyone was showing interest. That was something I like that everyone was taking it seriously and they had some questions in mind…"

Moreover, participants expressed concern over the limitations of using screen-based technology for learning self-defense, as one needs the involvement of other people to acquire the practice. Despite this, it is interesting to note that the sense of community and the role of the group were not that prominent in the speculative designs.

4.1.6 Self-Defense as Women's Knowledge

Throughout the discussions with the participants, there was a prevailing sense that the lessons shared during the self-defense class are seen as essential knowledge for existing in the world as a woman. Being prepared for possible situations of assault or harassment was seen as a survival skill and any gaps in this knowledge as security risks. Some participants communicated disappointment that this type of training even exists, however the knowledge gained was still seen as practically valuable.

> "This is a topic that all women should know about. Because you know yourself, but you don't know the people outside. So that's why you make yourself educated…"

Both the instructor and the other participants were seen as sources of knowledge, with the discussion session at the end of the workshop being seen as an opportunity to learn from the experiences of others. Participants also communicated uncertainty over remembering all the information shared during the workshop and a need to internalize the lessons. This was reflected in the discussions on the use of screen-based technology in self-defense, where participants noted the need for an information-rich application. This belief was echoed in the speculative designs via the imagining of a technology that can guide a person during a fighting sequence via voice interactions, as well as in the form of a personal AI chatbot that is specifically designed to educate women on the topic of self-defense and empowerment.

4.1.7 Felt Experiences as a Resource for Designing Interactions

The above presented can be seen as one half of the technosocial design material. As a resource for design, these felt experiences and emotions are fluid, constantly evolving, and often contradictory. They can be seen as a source and cause for movement, and as influenced by and influencing the class's environment. [31]

Working with emotions as materials presents itself as a specific challenge. However, certain qualities of movement can be seen as potential triggers for emotional states. As such, a system that aims to elicit a particular state may prompt a type of movement that is associated with said state [31], e.g. prompting a certain manner of walking to trigger feelings of strength. However, this should not be seen as a guaranteed result due to the subjectivity of each experience, and the fact that emotion is an emergent quality that exists in the space of the interaction between the user and the system [31]. Moreover, any design that situates itself in the space of self-defense should display an awareness that movements performed in this context are meaning laden. Namely, learning to adjust one's posture for the sake of having a "better" posture is experientially different from learning to adjust one's posture for the sake of avoiding assault, even though on a bodily level it may trigger the same muscle movements. It should also be noted that different movements may bring about different associations. Some participants experienced the act of shouting "No" and "Stop" as a purely physical activity, while others as a consequence of an emotional state. In the same manner, having an assertive stance was seen as both something the body has to learn and as an outcome of mental strength. Furthermore, concerning common beginner hurdles, there should be an awareness that even though a certain movement interaction is physically within-reach [37], it may be perceived as out-of-reach [37] due to it being outside a participant's movement repertoire.

4.2 Feminist Design Speculations

The design workshop resulted in a collection of concepts inspired by the partitioners' self-defense experiences. The designs are constructed in a way that provokes "to imagine them in use and the possible future [of women's empowerment] that would manifest around them" [62]. It is interesting to note that despite the predominant focus on the affective during the

interviews, the focus of the crafted speculations is mostly on the corporeal. This can be seen as demonstrating the inseparability of mind and body during the practice of feminist self-defense.

4.2.1 Mocking up reality – Technology as a facilitator of safe "unsafe" spaces



Figure 1[9] : Optimal Grip[10].

The design of "Optimal Grip" (Figure 1) consists of a bodysuit with built-in data gloves and a VR headset. The body suit is made of a textile that can expand, contract, or trigger thermal stimulation, to simulate contact from another body. The VR headset is imagined to be light and unobtrusive, so that it "disappears" in use. During the interaction, the user can switch between a first-point perspective and a third-point perspective. The latter can be initiated when additional help or advice is needed. As such, the third-point perspective can be seen as an observation of a demonstration of self-defense techniques that would be most appropriate in a given situation, while simultaneously, via the body suit, feeling the sensations that might arise. This builds on the notion that watching the performance of an action can be an important part of learning how to perform that action [35]. The sensations from the body suit mitigate the aspect that just observing a

9 The selection of the presented speculative designs is made by the researcher based on how evocative they are of each theme. Following the principle of "shared creativity" [20] some of the concepts are further expanded and visualized in order to make them more tangible to a viewer. The visualizations aim to ground the designs in the familiar, while at the same time making them unfamiliar enough so that they can be separated from the present and experienced as speculative concepts [3].

10 The visualization of "Optimal Grip" was driven by the question of how should a garment that one puts on to "experience" an attack look like. The crafted speculation moves away from existing visualizations of VR body suits and turns towards textile design for inspiration. The indented outcome is the rendering of a fabric with sections that mimic the curves of the body and a visible yet unrecognizable textural feel.

The concept visualization was created via a combined use of Midjourney - a generative AI software, and Adobe Photoshop CC.

performance of an action does not translate to knowledge of kinesthetic sensations such as pressure, pain, or the felt gravitational pull [35]. Interacting with the design can be seen as an exercise in using one's entire body, which as mentioned during the self-defense class, may not come naturally for most people. The workshop participants that presented this concept envisioned the interaction as an individual experience since:

> "In a real-life attack you are most likely to be alone"

Moreover, they stated that relying solely on a VR headset might result in a less immersive experience, as the interaction is mostly reduced to what the user can visually perceive.

This speculation is an amalgam of several similar concepts inspired by the "Pain" and "Group Dynamic" prompt (Appendix G, Prompt 2 and 3). The design speculates that engaging in a simulated attack scenario, without being in real danger, can lead to a better understanding of one's bodily reactions. As such, the concept can be seen as shaped by the perceived unpredictability of the soma, as one cannot know how one will act in a high stress situation. In this way, the design implies that the facilitated experience could further one's knowledge about oneself. A workshop participant explained this as boosting your evaluation about yourself by reflecting on your own expectations of an attack scenario, as well as the triggered bodily responses and feelings.

This concept can be seen as a response to the experienced separation between the safe environment of a class and the material reality of an actual attack and as pointing to a need to test one's newly gained skills, and further one's knowledge of self-defense. However, the speculation also raises further questions. Namely, how will the attack scenarios be rendered? How is the perpetrator presented? Even though the attack is not real, the emotions felt are real, and as such how should they be processed?

4.2.2 Technology as "training wheels" for gaining confidence



Figure 2. Fighting Baymax [11]

11 The visualization of the concept is driven by the question of how would a robot that is placed in a gym or a dojo that teaches feminist self-defense look like. It purposely moves away from more stereotypical images of futuristic robots, by

The aim of the robot (Figure 2) is to challenge the person interacting with it to exert more force during a mock-fight. It is made of a soft material and contains sensors that measure the impact of a hit or kick. It can then prompt the user to "hit harder" either via audio or colored light output. The robot's movements in and out of the user's peri- and extra-personal space and around its own central axis are prompted by the user's movement. On the other hand, the robot's speed and rhythm will affect the practitioner's movements, thus leading to a coupling between technology and human, a "complete somatic communication" [30]. When engaged for longer periods of time, the tactile feel of the surface of the robot changes, becoming warmer or even wet to the touch, to simulate the feeling of perspiration on skin.

This design was created in response to the prompt "Pain" (Appendix G, Prompt 3) and can also be seen as an exercise in using one's entire body. The workshop participants speculated that interacting with this technology would overcome the issue of beginners restraining strength during mock-fighting.

"You can practice hurting it, because it is not a human."



Figure 3. A room of one's own[12].

utilizing soft organic shapes and hiding most of the technology from view. Inspired by a participant's quote "you can practice hurting it, because it is not a human", its shape can be seen almost as an abstraction of the human form. The concept visualization was created via a combined use of Midjourney - a generative AI software, and Adobe Photoshop CC.

12 The visual depiction of "A room of one's own" aims to capture the imagined playful quality of the experience via the use of color and the organization of the interactive objects. Namely, the objects are scattered across the floor, placed on the walls, or suspended from the ceiling. They have ambiguous forms to not impose meaning and allow practitioners to decide how they wish to interact with them. The concept visualization was created via a combined use of Midjourney - a generative AI software, and Adobe Photoshop CC.

The concept in Figure 3 envisions a space where one can engage with one's physicality in a variety of ways, from practicing kicks and hits to the use of an assertive voice. The design does not prescribe a certain way of moving. Instead, it encourages participants to engage in movements that elicit joy. It thus prompts inquisitiveness about one's own physicality. Kicking, hitting, or stomping on the elements of the room is accompanied by the emission of colored lights, however it is up to the user to construct their possible meaning. The potential simultaneous use of movement and voice can be seen as a practice in voice-posture symmetry[13]. Depending on preferences, up-tempo music that inspires movement can be played in the background. This design was created in response to the "Pleasure" prompt (Appendix G, Prompt 4) and it was informed by the participants' view that to derive pleasure from the practice, one needs to also feel confident.

"Confidence is key to pleasure, to feel pleasure you need to be confident, and vice-versa".

They thus envisioned the room as a safe space where in an almost playful manner it can be possible for practitioners to test and push their own limits, as well as reflect on the experience. The workshop participants speculated that interacting with this design would allow them to derive pleasure and confidence from experiencing the improvement in one's skills.

The above presented concepts reflect the idea that a "significant change in body or voice might affect how we think" [35] and speculate on the possible role that technology can play in facilitating that change. They explore how different modes of movement may enact bodily change and further lead to gaining confidence, strength, and potentially reducing feelings of vulnerability. These designs were conceived as tools beginners rely on for a limited amount of time, until they feel confident in their skills. Or as one participant stated:

"Should you be dependent on technology, or should you be self-empowered?"

This implies that technology can play the role of a facilitator of self-sufficiency and embodied empowerment.

5 DISCUSSION

5.1 Feminist Self-defense as a Design Space: From Experiences to Interaction Design Qualities

The speculative designs open the design space of feminist self-defense by imagining technology that is built on its values. They are envisioned as technology that supports the teaching of a bodily practice [42] and can be used as a starting point for the design of future interactions. Based on the designs' goals and features, a set of 7 design qualities were derived that can guide future explorations in this space.

- **The "playful - serious" spectrum:** The experiential qualities of the concepts can be mapped along a spectrum between playfulness (as in "A room of one's own") and seriousness (as in "Optimal Grip"). Playful interactions were seen by the participants as a way to ease one into the practice. However, some also viewed this quality as a distraction from the main goal, which in turn necessitates the design of interactions that feel more serious.
- **Prompting movement:** The manner in which a system prompts action should also be a subject of attention, as moving in a way that is beyond one's habitus may elicit feelings of anxiety and discomfort. In the self-defense workshop, the instructor helped to overcome this, by stating the importance of an action (e.g. learning to shout)

13 A voice-posture symmetry refers to a lesson from the feminist self-defense workshop which indicated that to be more successful in verbal resistance one's use of voice (loud and deep), verbal articulation (precise), and posture (upright, assertive, maintaining eye contact) should all be in sync. An assertive use of voice that is not accompanied by an assertive body language may be less effective.

in defending oneself. Even though not explicitly addressed in the speculations, interaction designers may need to consider how a system might elicit (and sustain) movements that are outside of the feminine habitus.

- **Use of multiple modalities:** In the self-defense workshop, practicing some of the defense techniques can be done simultaneously, such as using an assertive voice and an assertive posture in parallel. This can be translated to the design of interactions that rely on a combination of modalities, such as the combination of movement and voice in "A room of one's own". Thus, designers may need to evaluate which modalities can and should be combined to better support the learning process.
- **Reflection-in-movement:** Except for an AI chatbot, all the concepts envisioned movement as the primary mode of getting to know one's soma and reflect on one's boundaries and inhibitions. Reflection was seen as an outcome of exploring one's physicality and as an important step in developing embodied knowledge. As such, designers may need to consider how reflection may be facilitated by different modes of movement, from more exploratory ones as in "A room of one's own" to more directed ones as in "Optimal Grip".
- **A non-quantified experience:** It is also interesting to note that none of the speculative designs seek to quantify the user's experience, nor to detect and identify emotional states. Thus, the designs do not frame the body as an object, but instead lead to the creation of "embodied subjectivity" [31].
- **Designing with emotions:** The interaction with the imagined technology may trigger emotions that are generally seen as "unlady-like" (e.g. aggression) and undesirable (e.g. fear). However, this is not done to learn to overcome or suppress them but to become familiar with them and act in their presence. In this way, the concepts support the notion that emotions are part of corporality, help the practitioners understand how different emotions may arise in different situations, and how they might influence their actions [31].
- **Kinesthetic empathy:** This phenomenon refers to how observation of movement may elicit bodily responses in the observer [21]. In the speculative designs, this can be seen in the third person perspective feature of "Optimal Grip". As such, designers should assess what role (if any) facilitating kinesthetic empathy could play in the interaction design.

It should be noted that the speculative designs frame the experience as more of an individual practice, which somewhat contradicts the importance of the group when learning self-defense.

5.2 Technology That Facilitates Embodied Empowerment

The existing research on feminist self-defense identifies "strength uncovering" as one of the key goals of the practice [56]. This is seen as a catalyst for developing self-sufficiency, challenging gender stereotypes, and ultimately empowering the practitioners [56]. In both this study and existing literature, embodied strength is conceptualized as stemming from the development of embodied skills. As such, the body and tacit knowledge is where strength and by extension empowerment originates. The speculative designs thus envision technology that can be used as a tool to develop such skills. They play the role of tools for learning self-defense, as opposed to tools for enacting self-defense (e.g. weapons). Focusing on the design of the latter might not be the optimal way of supporting the practice, as feminist self-defense is much broader than the teaching of how to use weapons. As discussed by self-defense instructors, a more effective solution is knowing how to think on one's feet and learning how to use common objects (a bag or a jacket), the affordance of the environment (a well-lit street, a row of parked cars[14]) and of course one's body (voice and posture) as "tools" for deterring an assault. However,

[14] When one is walking alone or if feeling that they are being followed, a possible course of action is to take a route that passes along a row of parked cars. If needed one can activate a car alarm(s) on purpose in order to draw attention to an unfolding unsafe situation.

to be able to perceive and use such affordances, one needs to embody self-defense skills. From a phenomenological perspective, this can be viewed as a skillful interaction with one's lifeworld [61] which frames the speculations as technology that helps support said interaction [61]. Additionally, as through the lens of embodied cognition, the purpose of tool design is to help overcome the physical limitations and extend the abilities of the body [35], the speculative designs can thus be seen as tools that reshape the perception of what a female gendered body is capable of.

Moreover, considering that gaining skills in self-defense was seen as gaining knowledge, the speculative designs can contribute to women's empowerment by facilitating access to informational resources - how one is able to defend oneself, knowledge about one's body and emotions, and ideological resources - challenging sexed embodiment via transgressing the feminine habitus. In this manner the designs and the emotions that can potentially be elicited from interacting with them, can be seen as scaffolds in developing embodied skills and cultivating embodied empowerment - the "new soma reality".

5.3 Speculating with the Body

The speculative designs are imagined as tools that help women learn that they are indeed capable of self-defense - thus, undermining sexed embodiment and challenging the status quo. The concepts also point to a certain preferable future, where technology and design can play a role in challenging gender ideals. Namely, as the speculative designs help teach women how to be able to defend themselves, they imply a belief that women's bodies are not inherently weaker than men's bodies. Since design artifacts reflect social values and contribute to their solidification [4], the created speculative designs can be seen as contributing to a subverting of gender norms. They do not frame the female body as an object, nor as a reflexive project [43], but as a source of agency and "an agent for social change" [43].

The provocation of the speculations thus may not lie in their technological features, but in the way that technology is utilized. Namely, it implies a future state where technology developers do not neglect the needs of female gendered bodies [14]. On the contrary, it imagines how technology and design, by supporting women in connecting with their bodies and emotions, can be vehicles for empowerment and liberation. Beyond the space of feminist self-defense, these speculations can be used as a prompt to reflect on the design of embodied interactions for corporeal practices (e.g. athletics, dance, martial arts) and looking more critically at how women's bodies, their needs, and abilities are framed, and which values (implicit or explicit) guide the design of the technology. This can be seen as contributing to the work of [57].

The participatory speculative design process itself can also be viewed as empowering, as it created a space that enabled the participants to envision feminist technology based on their soma experiences, and to further reflect and process said experiences. Additionally, as design methods can be used as tools for understanding how we make sense of the world [40], the process of design speculation can be seen as a tool for gaining insight into the subjective experience of the self-defense practice. Moreover, seeing the speculative designs as materialization of felt emotions - "the qualitative dimension of an experience" [40], indicates that this method may be used for uncovering essences of bodily experiences.

5.4 Beyond the Speculation

The imagining of a future where the presented concepts are seen as obsolete technology was touched upon during the design workshop. The larger context in which the designs exist was also briefly explored. Namely, if they are part of a capitalist society in which the matrix of domination [48] exists - who will have access to and benefit from the imagined future state of technological development, what their environmental impact would be, how might they be used by bodies of different abilities and sensory needs, how should users' data be protected, as well as possible misuses or malfunctions of the designs.

Moreover, the debate on whether feminist self-defense leads to individual or collective empowerment [43] and the question of the necessity of simulated volatile experiences[15] [43] can extend to technology that aims to support the practice. However, currently, these questions are beyond the project's scope.

5.5 Future Research

Future studies could invite self-defense teachers and other soma-practitioners to take part in the speculation. Moreover, as currently the designs do not explicitly engage with any of the more volatile emotions that can be triggered during the practice, future research that explores this aspect can also be designed.

6 CONCLUSION

The aim of the project was an exploration of a novel design space - feminist self-defense. Grounded in theories of embodiment, the study first inquired into the emotions and experiences of first-time participants via semi-structured interviews. This was followed by a participatory speculative design workshop, in which the participants were prompted to use their embodied experiences as a resource for imagining the design of technology that would support their practice. This resulted in several speculative design concepts that point towards technology playing the role of a tool for gaining knowledge about one's soma and as a learning tool for acquiring self-defense skills. They thus empower the participants by facilitating access to both informational and ideological resources. Analysis of the interviews and the speculative concepts lead to the identification of 7 design qualities which can be used as a guide for further research in this design space.

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15 Some self-defense programs contest the use of mock-assault scenarios due to the belief that it may lead to further trauma and question their efficacy in empowering women. [43]

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A APPENDICES

A. Interview Questions for Self-Defense Instructors

Intro:
Introduce ourselves
Introduce what we do
Introduce how the data and insights will be used
Ask for consent
Inform of freedom to not answer a particular question(s)

Warm-up:
For how long have you been an instructor?
Why did you become a self-defense instructor?
What's the best part of being an instructor?

Interview:
What type of self-defense do you teach?
What are the goals of a self-defense class?
What does the process look like when teaching women's self-defense? Can you walk us through step by step?
How (if at all) do you include feminism in the course?
How are mental and emotional aspects discussed in the course?
How is empowerment discussed?
How are boundaries taught or discussed?
How do you address fears and anxieties that participants may have when it comes to self-defense?
How do you navigate the balance between teaching practical self-defense skills and avoiding victim-blaming rhetoric?
Can you discuss the role of physical techniques versus verbal assertiveness in your training?
Are there specific strategies or techniques you teach for dealing with different types of aggressors (e.g., strangers versus acquaintances)?
How are the role play scenarios "designed"?
How do you create a supportive and empowering atmosphere in your self-defense classes?
Can you discuss the role of group dynamics and peer support in the learning process?
What are the challenges during the learning sessions?
Can you tell us a bit more about the motivations of the participants?
How (if at all) can you spot when a participant is struggling?
How do you address the emotional aspects of self-defense training, particularly in relation to trauma or past experiences of violence?
How (if at all) do you define that a participant has successfully completed the training?
Can you share any insights or observations about how participants' attitudes towards themselves and their bodies change throughout the course of the training?

Additional Questions:

Do you have any material like survey results or PowerPoints that we can read?

Do you know any instructors or women that have done a self-defense course that we could contact for this research?

Close the interview:

Questions about us

Thank them for the time

Ask them if there are any activities in the future e.g. workshops and would they be open to us participating in them

Inform them that the project ends in June, and all insights can be shared with them afterward

B. Interview Questions for Self-Defense Students

Intro:

Introduce what we do

How the data/insights will be used

The possibility that quotes may be used from the interview, but if they are used it will be in a way that does not reveal the identity of the interviewee

Ask for consent

Freedom to not answer a particular question(s)

Warmup

Have you been in a women's self-defense class before?

Have you been doing any type of Martial arts before?

Questions:

What were your expectations prior to the self-defense workshops?

How did you feel before the workshop? (Eg. Anxious, excited…)

What part of the workshop stood out the most to you and why?

What aspects of the workshop did you find difficult?

Which lessons did you find the most valuable?

Can you reflect on any emotions that you felt that arose during the workshop?

Were there any moments during the workshop that challenged you mentally or emotionally?

How did it feel when you were asked to close your eyes and imagine being approached on the street by a stranger?

How did it feel when you were asked to take on a strong body posture and yell "No" or "Stop"?

During the role-playing sessions, how did it feel when you were asked to play the role of the perpetrator (try to grab or stab your practice partner)?

How did you feel about the aspects of the workshops that addressed women's experiences? Eg, the fact that we normally would not take a walk through a park at night even if it meant getting home faster (We can make this question more or less specific)

How would you describe the atmosphere during the workshop?

How did you find the discussion session at the end of the workshop?

After the workshop, what are your thoughts on your body's physical ability?
What other topics/skills/discussions would you have liked to be a part of the self-defense training?
What changes (if any) have you noticed in your self after the workshop?
Have you noticed any changes in how you experience the environment when you exit your house or go outside at night?
Did the workshop change your perception of personal safety or risk assessment in any way? If so, how?
In what ways do you think attending this workshop might impact your confidence and sense of empowerment moving forward?
If you would teach self-defense to someone else, what do you think is valuable to teach?
Imagine that an app for self-defense is to be built, what do you think is important to include in the app?

Wrap up:
Ask them if they want to add something else about the experience that we might not have asked
Ask if they have any questions
Thank them for their participation

C. Workshop 1 Consent Form

Information about participation in research project for women's self-defense

Information about the project
This project is part of two master theses on the department of Informatics and media at Uppsala University. The theses are about bodily movements and experiences in feminist self-defense classes. The purpose is to bring knowledge about feminist self-defense (the process, bodily and emotional experiences, and movements) to the area of technology and design. You are asked to participate in this workshop, the following interview, and a second workshop to contribute to the data collection of the master theses. All of the participants are women because of the chosen case - women's self-defense.

What participation in the study would involve for you
The workshop will be two hours and the following interview will be approximately 30-60 minutes. During the workshop you will be observed by the researchers. The workshop will be filmed, with the sole purpose that the researchers will look at the video material afterwards to have a closer look at movements.

If you in any way feel uncomfortable physically or mentally before, during or after the workshop, please contact the researchers as soon as possible. You are fully allowed to end your participation whenever you want and this will not affect neither you or the research.

How to learn about study results
You will be able to partake of the results of the study when the project is done (in June). If you would like to get access to the theses, please contact the researchers.

Participation is voluntary
Participation in the project is completely voluntary. At any time, you can choose to no longer participate and you do not have to explain why. If you choose to no longer participate, this will not affect the research in a negative way. If you no

longer wish to participate, you must notify the persons in charge of the project. If you wish for your data to no longer be part of the study, you can revoke your consent at any time by contacting the thesis researchers.

Contact information to those responsible for the project
Rosanna Moberg - rosannamoberg@gmail.com, 0768616264
Bojana Nikolovska - b.b.nikolovska@gmail.com, +46 70 286 20 62

Consent to participating in the master thesis research project on women's self-defense
I have read and understood the information about the study. I have been given the opportunity to ask questions and I have had them answered. I may keep the written information.

☐ I consent to participating in the study.
☐ I consent to being filmed during the workshop.

Place and date	Signature and clarification of signature
……	……

D. Workshop 2 Consent Form

Information about participation in a speculative participatory design workshop in a thesis project that focuses on women's self-defense

Information about the project
This project is part of a Human-Computer Interaction master's thesis in the Department of Informatics and Media at Uppsala University. The thesis is about bodily movements and experiences in feminist self-defense classes. The purpose is to bring knowledge about feminist self-defense (the process, bodily and emotional experiences, and movements) to the area of technology and design.

What participation in the study would involve for you
The workshop will be two hours and during it, you will be asked to speculate on the topic of possible design of technology that would fit in the practice of feminist self-defense. The design speculations will be presented as part of the final thesis project in either the original or modified version. The workshop will be filmed for the purposes of documenting the speculative design process.

How to learn about study results
You will be able to partake of the results of the study when the project is done (in June). If you would like to get access to the theses, please contact the researcher.

Participation is voluntary

Participation in the project is completely voluntary. At any time, you can choose to no longer participate and you do not have to explain why. If you choose to no longer participate, this will not affect the research in a negative way. If you no longer wish to participate, you must notify the person who moderates the workshop. If you wish for your data to no longer be part of the study, you can revoke your consent at any time by contacting the thesis researcher.

Contact information
Bojana Nikolovska - b.b.nikolovska@gmail.com, +46 70 286 20 62

Consent to participating in the master thesis design workshop on women's self-defense
I have read and understood the information about the study. I have been given the opportunity to ask questions and I have had them answered. I may keep the written information.

☐ I consent to participating in the study.
☐ I consent to being filmed during the workshop.

Place and date	Signature and clarification of signature
……	……

E. Workshop 2 Protocol

Feminist self-defense - Participatory Speculative Design Workshop
Duration: 2 hours 13:15-15:15

Preparation
13:00 - 13:15 Set up the room

Part 1
13: 15 - 13: 20 Welcome participants and signing of consent forms

- Explain the goal of the thesis
- Explain the goal of the workshop - creative speculative designs that would fit in the practice of feminist self-defense, using design as a way to imagine what could be
- Emphasize that this is a safe space where they can reflect and explore their experiences
- Present the agenda
- Instructions for the participants: when creating the prototypes, not to focus on what is feasible, what would make sense as a commercial product, or how exactly the technology would work, suspend your disbelief, and focus on imagining things you would like to exist while using your own experiences and emotions as material/inspiration
- Divide themselves into groups of two or 3 participants

13:20 - 13:30 Physical warm-up

- Head rotation
- Shoulders rotation

- Arms rotation
- Side stretch
- Side swing of arms (start slowly, speed up, then slow down again)
- The participants will be prompted to do the exercises slowly and to focus on how different movements feel in the body. To focus where they might feel tension or if anything is making them uncomfortable. Emphasize that this part will not be filmed so they feel more comfortable
- Emphasize that it's okay if this feels a little silly and that that is the point. As we sometimes get focused on how our bodies should and shouldn't move, and anything outside of the repertoire of movements feels uncomfortable to us.

13:30 - 13:33 Relaxation exercise
Participants will be prompted to do a yoga body scan and focus on their breathing

13:33 - 13:43 Body maps and discussing the body maps
Explain what body maps are and why we are doing them (a way to externalize feelings that are sometimes difficult to explain just verbally, they can be as specific or abstract in their maps as they wish to, and they can use a variety of crafting materials to fill in the map and they can be as rough with the paper as they wish to)

Making the body maps 5 minutes
Discussing the body maps with their group partner for 5 minutes

13: 43 - 14:00 Body props
- Introduce the body props as objects to imagine with
- They elicit different sensations depending on which part of the body they are placed on
- It is up to the participants to decide what they mean or what they do
- Prompt them to select one body prop for the group, take turns exploring how it should be placed on the body, and try to do some of the different exercises we did during the self-defense workshop. (E.g. Stand like Mike Tyson, try the hand grabbing exercise, or defending from stabbing exercise)
- Emphasize the needed mindset: exploratory mindset, playfulness, exploring body sensations, this type of imagining comes naturally to kids, e.g. they can look at a cardboard box and see armor in it or a robot's body. So, this exercise will be tapping into that imaginative thinking.

14:00 - 14:10 Prompt introduction and brainstorming start
- Each participant group will be given a prompt around which they can speculate and will be asked to imagine a future design that responds to that prompt. It will be explained that they can use the body props, (Eg. Imagine if that were a device what would it do) but that they are free to go beyond them
- As further support, some embodied interaction design frameworks will be presented (as a slide): tech that is worn on the body and that is able to extend the abilities of the person and is essentially an extension of the person, tech that is able to connect people and facilitate close collaboration, ambient technology or "calm technology" that exists in the immediate environment/ in the background

Break 10 minutes

Part 2

14:20 - 15:00 Prototyping continues

15:00 - 15:10 Presentations

15:00 - 15:15 Closing words

- Ask if anyone has any questions or anything to share
- Thank the participants for their time and effort
- And let them know that the results will be shared with them

15:15 - 15:30 Tidy up the room, and collect the materials

F. Body Props



Figure 4 The four body props were made from cotton fabric with filling made of synthetic fibers or plastic granules. Marbles were added to the filling of two of the props in order to create a variation of weight among the objects. Satin ribbons were attached to the objects in order to allow for flexibility in object manipulation.

G. Speculative Prompts

Prompt 1: Breath

> "We learn to yell the techniques we use ("KICK!" "KNEE! " "GROIN! " "EYES!") or to simply yell "NO! " as we fight. This is to ensure that we continue to breathe during the fights, that we intimidate our assailant, and that we get the attention of anyone in the area."
> Real Knockouts: The Physical Feminism of Women's Self-Defense, Martha McCaughey, pg.62

The importance of breath for the practice of feminist self-defense can not be understated. It is what makes us able to use our strong voice and shout. Having control over our breath can even influence our reaction time in certain situations. However, when we are startled, when we are in a stressful, tense situation, our breathing can become shallow. During self-defense training, teachers often have to remind the students to breathe during the execution of certain self-defense movements or during role-play situations. So, what if we designed technology from this standpoint?

Reflect on your experience of how it feels to be out of breath (eg. during a strenuous exercise), as well as how your breathing feels when you are stressed, anxious, or scared. Recall the breathing exercise we did during the self-defense class - using our breath in order to make our voice louder and more powerful.

How can we be inspired by this in order to design technology that will fit into the practice of feminist self-defense?"

Prompt 2: Group Dynamic

> "It was easier to see other women increasing their skills. I was still in my head saying, "I can't do this, I can't do this. But you know, so and so is doing it. Whoa, she knocked [the mugger] down in three shots. She is doing it." And that made it easier to judge my own progress."
>
> "As soon as someone was having a hard time, I had a hard time. . . . When they would start crying, I would start crying immediately and just be in an awful lot of pain with them. . . . What they were going through, I was going through, I couldn't separate it."
>
> Participant statements from the study "The role of the group in acquiring self-defense skills. Results of a qualitative study" by Fraser and Russell

Feminist self-defense teachers use a variety of mechanisms to facilitate trust, group cohesiveness, and bonding. As such, participants report that they view other self-defense practitioners as a support mechanism. They often feel like the group is a safe space where they can share and process past traumatic experiences. Moreover, for many practitioners, this is the first instance when they see other women exhibit strength, protect their boundaries, and successfully defend themselves from a mock attack. This in turn serves as inspiration and motivation for them to continue their practice. In addition, experiences such as these can be a catalyst for challenging conventions about what it means to move, act, and exist as a woman. So, what if we designed technology that took all this into consideration?
Reflect on your own experience of being part of a group during the self-defense workshop. How did it feel when we were all walking together and shouting, when we were asked to practice getting up from the ground, when we were sparring, or during the discussion session at the end of the workshop?
How can we be inspired by this in order to design technology that will fit into the practice of feminist self-defense?

Prompt 3: Pain

> "Abuse and violence can progressively induce in women a state of boundarylessness or "ownability""
> Real Knockouts: The Physical Feminism of Women's Self-Defense, Martha McCaughey, pg.9
>
> "Women aren't supposed to fight and aren't supposed to hit back and they're supposed to be nice and sweet, and not say no, and not do anything to tick people off. And to get women to go through that and actually fight and strike after years and years of being brought up, you know, where you're not supposed to, getting them through that "I don't want to fight" or "I'm not supposed to fight" [is the biggest barrier that women have to get over]. And once that's broken down, then everything else is pretty much easy sailing."
> Excerpt from The Fighting Spirit by Martha McCaughey, an ethnographic research on women's self-defense training

For some women, the practice of self-defense can be the first time when they are actively and consciously thinking about what it would feel and mean to be attacked. It also might be the first time where they experience what it means to "take a

hit", and in return inflict violence on someone else. For some practitioners, the classes can bring up negative emotions from past traumatic experiences. As such, during the training, it is not uncommon for practitioners to be triggered by something that they did not even know was a trigger for them, causing them to freeze or panic.
Reflect on your own experience during the self-defense workshop. What emotions did it raise in you? Did it prompt you to reflect on your own past experience?
Considering this, how might we approach the design of technology that will fit into the practice of feminist self-defense?

Prompt 4: Pleasure

> "When I was in one of my first self-defense courses, I heard a more experienced woman remark that she really enjoyed the sparring that took place in the mock attacks. I was stunned. Can we, should we, actually enjoy mock combat?"
>
> "The physical aspect is such a little part of what self-defense is. We women are always defending ourselves all the time, just the little slurs on the street, the looks, the stares, interacting with a boss or teacher that's not taking you seriously. That's all self-defense. That's the harder part of self-defense"
> Excerpts from The Fighting Spirit by Martha McCaughey, an ethnographic research on women's self-defense training

The practice of feminist self-defense can be often associated with negative emotions and experiences. During it, women often grapple with the reality of what it means to be attacked and they practice mock attacks which can vary in their level of violence. It is not uncommon for the classes to trigger memories of past negative experiences. However, despite this, many practitioners report deriving pleasure from the practice. This can be in the form of feeling stronger in their own body, confidence that if something were to happen they would be able to handle themselves. In addition, they often share how this newfound sense of strength has echoed in other areas of their life. Examples of this are being more confident in defending their boundaries at work or with romantic partners, having sexual confidence, having a better body image, and feeling overall more assertive in their everyday life. So, what if we designed technology that took all this into consideration?
Reflect back on your experience during the self-defense workshop. Did it elicit any feelings of strength, confidence, or empowerment? What triggered these emotions?
How can we be inspired by this in order to design technology that will fit into the practice of feminist self-defense?

H. Self-Defense Workshop Outline

- Introduction of sexual attack statistics
- Explanation of Swedish law on the right to defend on self
- Explanation of the components of a treat scenario (verbal or physical)
- Explanation of awareness skills
- How one can use the immediate environment as way to enact self-defense
- Reducing one's perceived vulnerability -" how not to be seen as an easy target"
- Exercise of assertive posture and walk
- Exercise in the use of assertive voice, shouting
- Use of common items for self defense
- Overview of courses of action in the case of an attempted robbery

- Self-defense in case of an attack with a sharp object
- Self-defense in case of an attack with a blunt object
- Self-defense in case of an attack with firearms
- Mental exercise of an imagined attack scenario
- Self-defense in case of a sexual attack
- Overview of techniques for releasing oneself from a hand grab
- Discussion on interrupting the script of an assault
- Discussion on courses of action in case of a completion of an attack
- Open discussion with all of the participants